Finite Element Approximation of Convection-Diffusion Equation

Vanishing Diffusion Limit and SUPG Stabilization

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Outline of the talk

- Model Problem: 1D steady-state convection-diffusion equation.
- Weak formulation and standard Galerkin FEM.
- Well-posedness
- SUPG stabilization method and its well-posedness.
- Error analysis of SUPG.
- Test Problem: 1D boundary layer.
- Numerical results comparing Galerkin and SUPG.
- Conclusion and future work.

Model Problem

Consider the steady-state 1D convection-diffusion equation:

$$-\epsilon \Delta u(x) + \mathbf{b} \cdot \nabla u(x) = f(x), \quad x \in (0,1) \tag{1}$$

with Dirichlet boundary conditions:

$$u(0) = 0, \quad u(1) = 0 \tag{2}$$

- ϵ: Diffusion coefficient $(0 < \epsilon < 1)$
- $\mathbf{b}$: convection field.
- $f \in L^2(0,1)$.

Weak Formulation

Find $u \in V = H_0^1(0,1)$ such that $a(u,v) = l(v) \quad \forall v \in V$.

Bilinear Form:

$$a(u,v) = \epsilon(\nabla u, \nabla v) + (\mathbf{b} \cdot \nabla u, v)$$

Linear Functional:

$$l(v) = (f, v)$$

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Discrete space:

$$V_h = \{v_h \in C[0,1] \mid v_h|_{I_i} = P_1(I_i)\ \forall i,\ v_h(0) = 0 = v_h(1)\} \subset V.$$

Well-posedneses: Lax-Milgram

Well-posedness is guaranteed if $a(\cdot,\cdot)$ is bounded and coercive.

- For coercivity: $a(v,v) = \epsilon||\nabla v||^2_{L^2}$.
- Using Poincaré inequality, we find $a(v,v) \geq \alpha\|v\|^2_{L^2(0,1)}$ where $\alpha = \frac{\epsilon}{C^2}$.

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Critical Failure as $\epsilon \to 0$:

- The coercivity constant α is directly proportional to ϵ.
- As $\epsilon \to 0$, the bilinear form loses its coercive property.
- Stability estimate $||u||_V \leq \frac{1}{\alpha}||f||_{L^2}$ "blow up" leading to massive spurious oscillations.

SUPG: Streamline-Upwind Petrov-Galerkin

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Stabilized Formulation:

$$a_h(u_h, v_h) + \sum_i (R(u_h), \tau_i \mathbf{b} \cdot \nabla v_h)_{I_i} = (f, v_h) \tag{3}$$

where $R(u_h) = -\epsilon \Delta u_h + \mathbf{b} \cdot \nabla u_h - f$ is the strong residual.

- τ is the stabilization parameter tuned to the Péclet number.
- SUPG is **consistent**: the exact solution u satisfies the equation since $R(u) = 0$.

Well-Posedness of SUPG

Streamline-Diffusion norm:

$$\boxed{|||v|||_{SD} := \left(\epsilon \|\nabla v\|^2_{L^2(0,1)} + \sum_{I_i} \tau_i \|\mathbf{b} \cdot \nabla v\|^2_{L^2(I_i)} + \omega \|v\|^2_{L^2(I_i)} \right)^{1/2}} \quad (4)$$

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We can not have divergence-free field and reaction coefficient to be 0 simultaneously. Atleast, this analysis will fail!!!

Local Inverse Inequality

Lemma (Local Inverse Inequality)

For the discrete finite element space V_h, there exists a constant $\mu_{inv} > 0$, independent of the element I_i and the local mesh size h_{I_i}, such that

$$\|\Delta v_h\|_{L^2(I_i)} \leq \mu_{inv} h_{I_i}^{-1} |v_h|_{H^1(I_i)} \quad \forall v_h \in V_h$$

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Lemma

Let the SD parameter τ_i satisfy

$$0 < \tau_i \leq \frac{1}{2} \frac{h_{I_i}^2}{\epsilon \mu_{inv}^2}$$

for each i where I_i is the ith element. Then the bilinear form is coercive, i.e.

$$a_h(v_h, v_h) \geq \frac{1}{2} |||v_h|||_{SD}^2 \qquad \forall v_h \in V_h.$$

Stability estimate: SUPG

Proof.

$$a_{SUPG}(v_h, v_h) = \epsilon \|\nabla v_h\|_{L^2}^2 + (\mathbf{b} \cdot \nabla v_h, v_h) - \sum_i \tau_i (\epsilon \Delta v_h, \mathbf{b} \cdot \nabla v_h)_{I_i} + \sum_i \tau_i \|\mathbf{b} \cdot \nabla v_h\|_{I_i}^2$$

$b \cdot \nabla(v_h^2) = 2v_h(b \cdot \nabla v_h)$ and $\nabla \cdot (bv_h^2) = (\nabla \cdot b)v_h^2 + b \cdot \nabla(v_h^2)$ □

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Stability estimate:

$$\boxed{|||u|||_{SD} \leq C \left(\|f\|_{L^2(0,1)}^2 + \sum_{I_i} \tau_i \|f\|_{L^2(I_i)}^2 \right)^{1/2}}$$

C is independent of ϵ.

Stability estimate: SUPG

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Proof.

Cauchy-Schwartz and $\|u\|_{L^2} \leq C|||u|||_{SD}$. □

Error Analysis: SUPG

Theorem (1)

Let the hypotheses of Lemma 2 be satisfied. Choose τ_i such that

$$\tau_i = \begin{cases} \delta_0 h_{I_i} & Pe_{I_i} > 1 \quad \textit{convection-dominated case,} \\ \delta_1 h_{I_i}^2/\epsilon & Pe_{I_i} \leq 1 \quad \textit{diffusion-dominated case,} \end{cases}$$

with appropriate positive constants δ_0 and δ_1. Then the solution u_h of the SDFEM satisfies the global error estimate

$$|||u - u_h|||_{SD} \leq C(\epsilon^{1/2} + h^{1/2})h^k |u|_{H^{k+1}}.$$

The *local mesh Péclet number* is defined as:

$$Pe_{I_i} := \frac{\|b\|_{L^\infty(I_i)} h_{I_i}}{2\epsilon} = \frac{\text{Advection Rate}}{\text{Diffusion Rate}}$$

The Diffusion Limit $\epsilon \to 0$

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Problem: *1D Boundary Layer*

$$\begin{aligned} -\varepsilon u''(x) + u'(x) &= 0, \quad x \in (0,1), \\ u(0) &= 0, \\ u(1) &= 1. \end{aligned}$$

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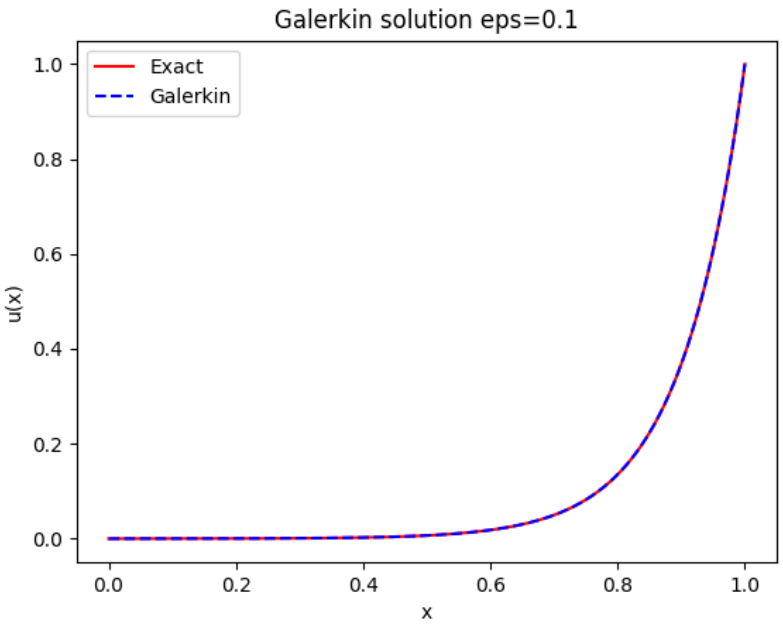
Exact solution:

$$u(x) = \frac{e^{\frac{x-1}{\varepsilon}} - e^{-\frac{1}{\varepsilon}}}{1 - e^{-\frac{1}{\varepsilon}}}.$$

Test 1: 1D Boundary Layer — Galerkin



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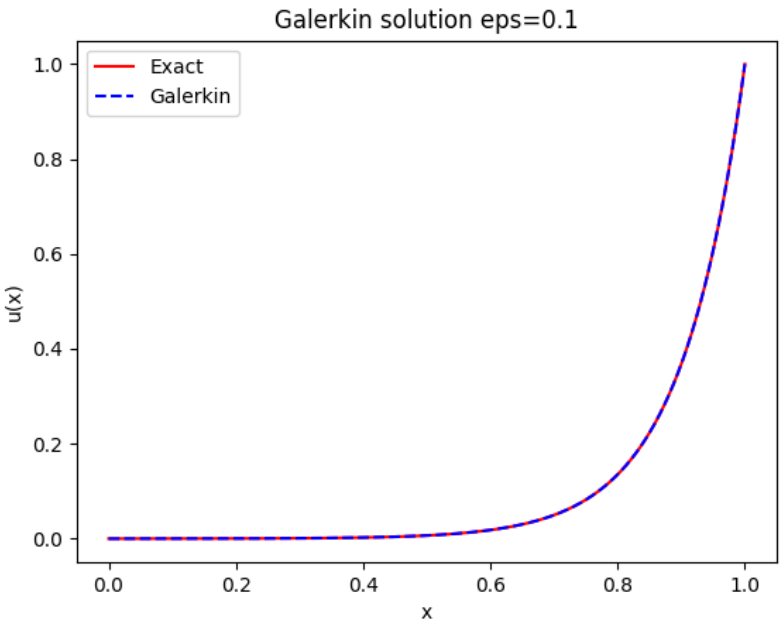

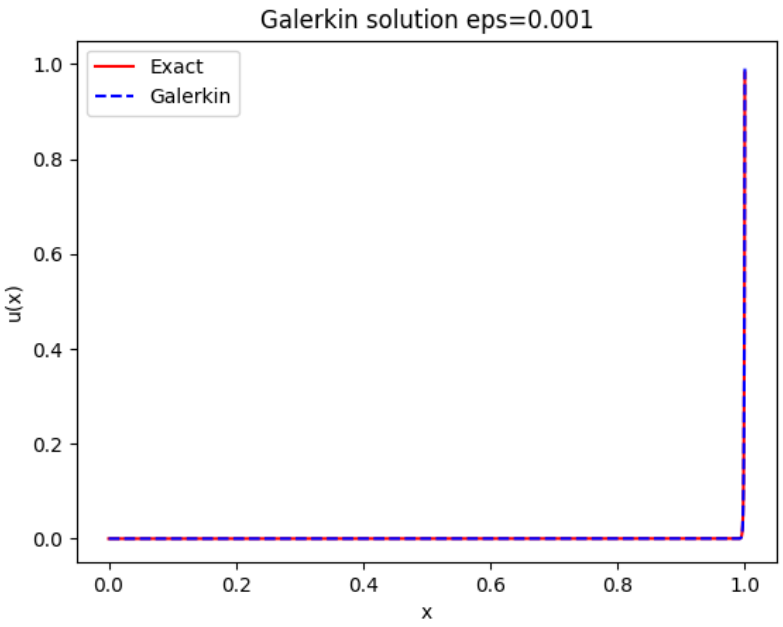

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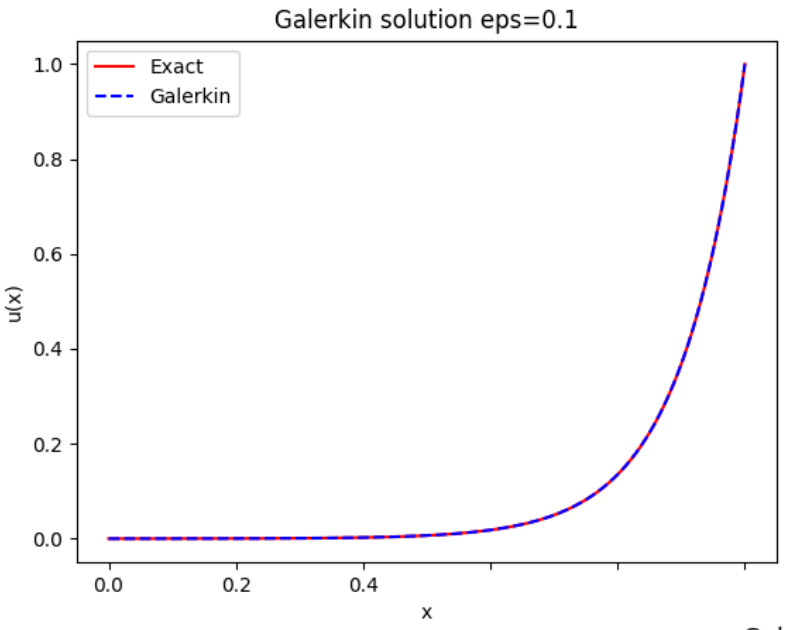

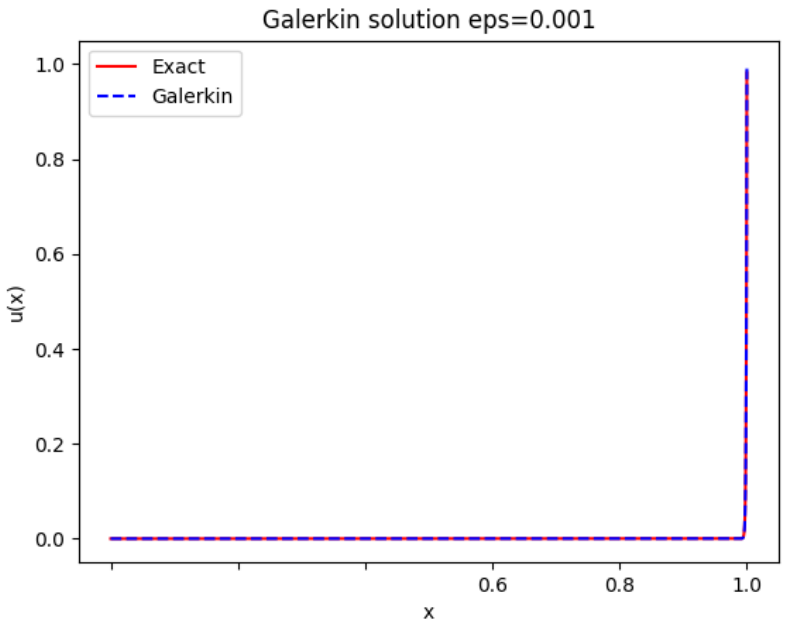

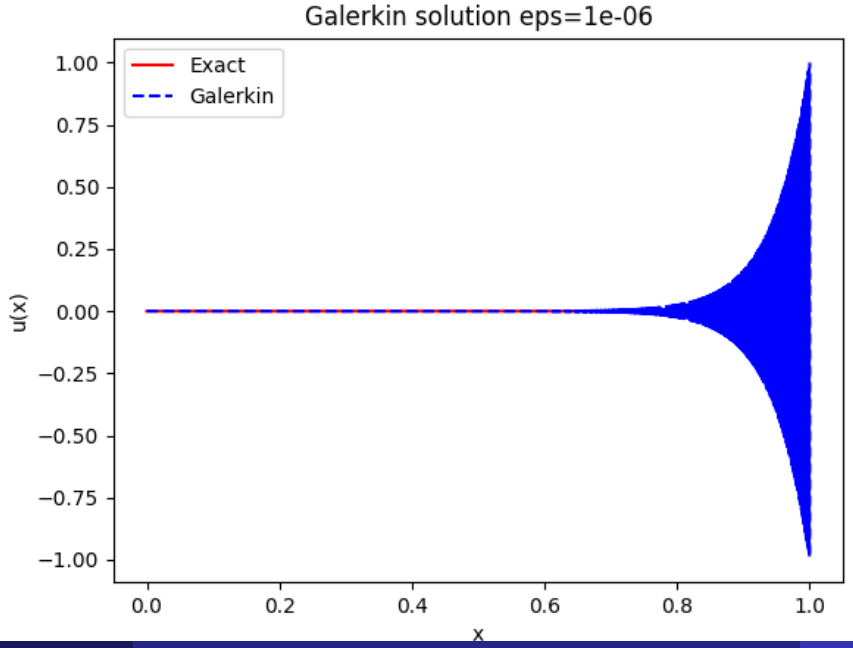

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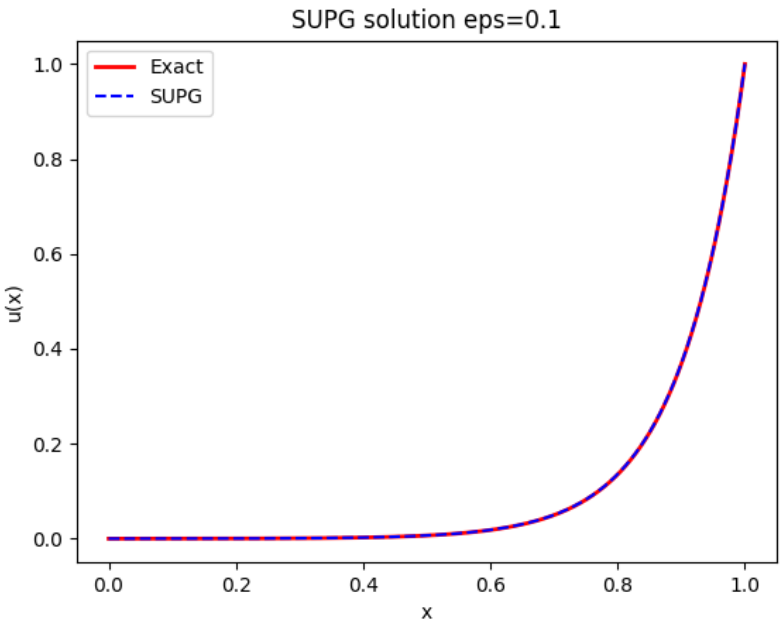

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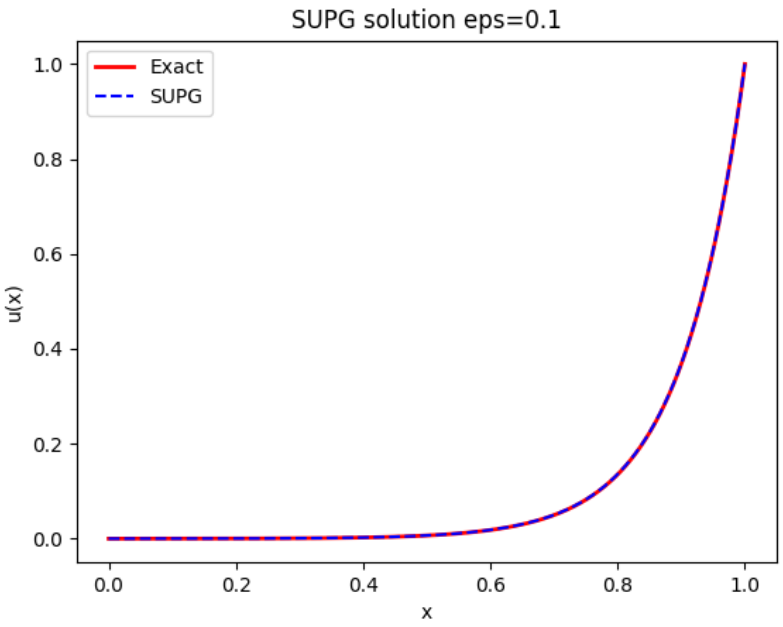

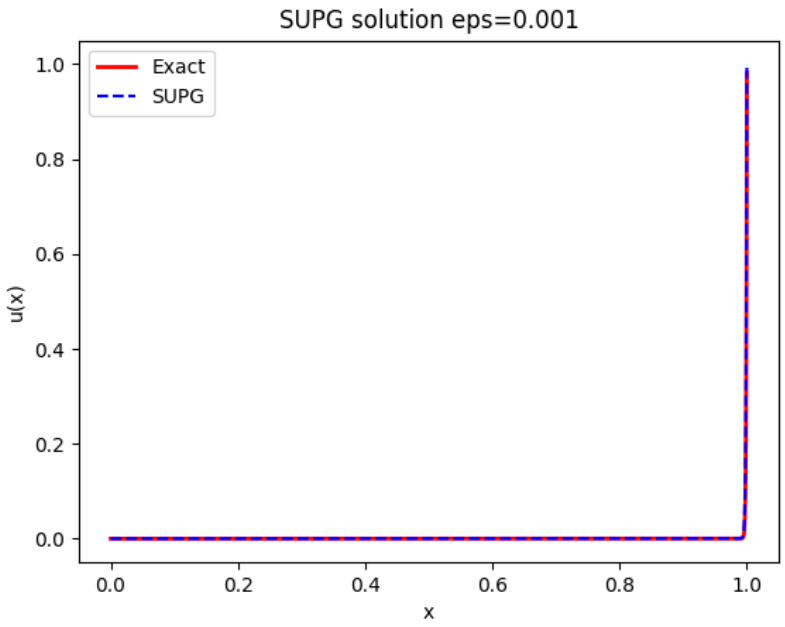

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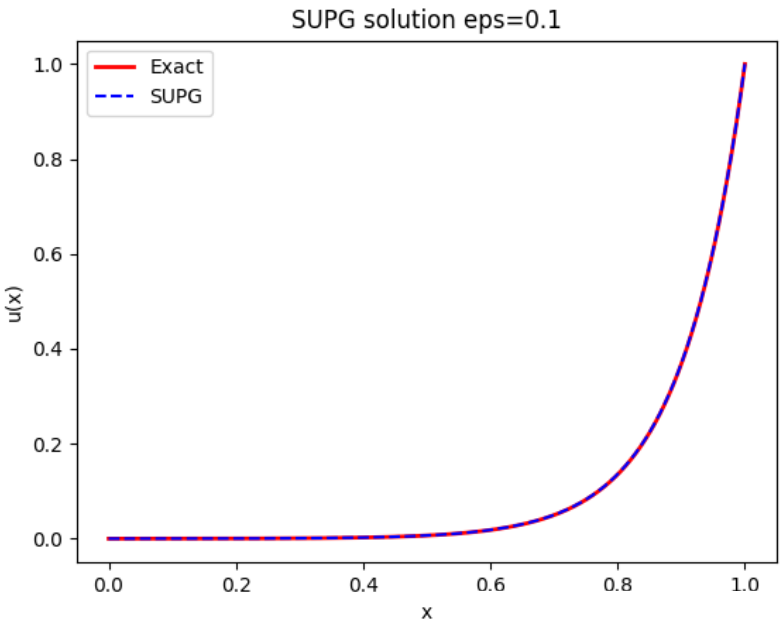

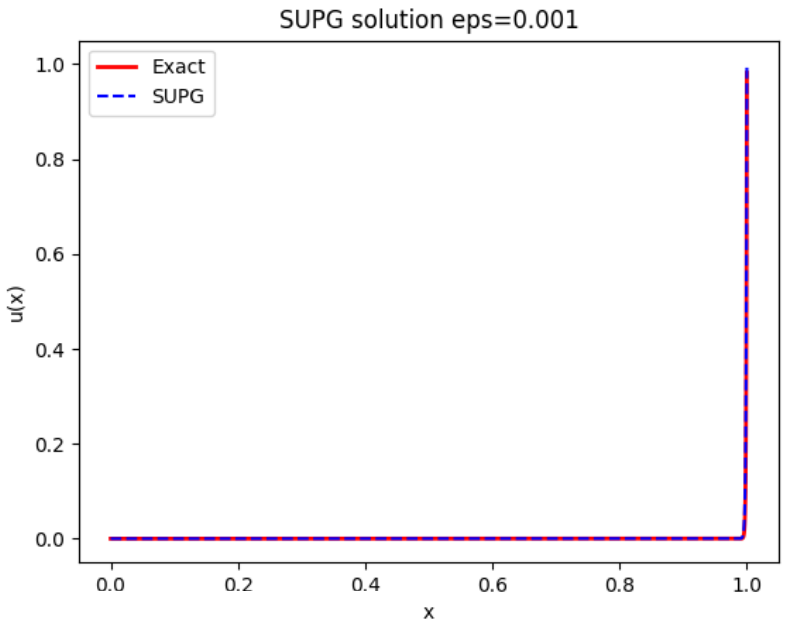

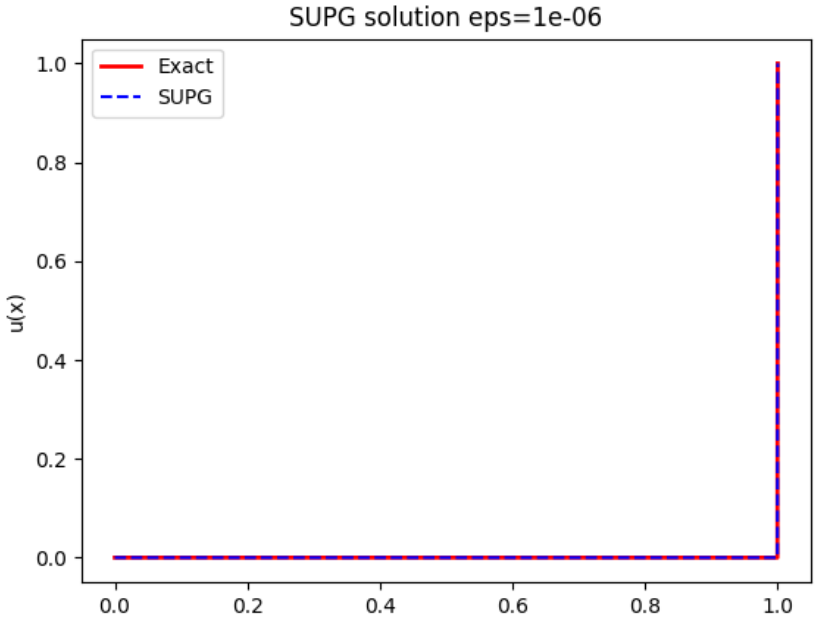

Error Comparison ($h = 0.0005$)

ϵ	Method	$\|\|u - u_h\|\|_{L^2}$	$\|u - u_h\|_{H^1}$	RoC L^2	RoC H^1
10^{-1}	Galerkin	3.89e-7	3.22e-3	1.99	0.99
10^{-3}	Galerkin	3.86e-4	3.2042	1.93	0.93
10^{-6}	Galerkin	0.1012	713.613	1.50	0.28

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10^{-1}	SUPG	3.94e-4	5.10e-3	–	–
10^{-3}	SUPG	3.75e-3	4.385	–	–
10^{-6}	SUPG	1.15e-2	123.33	–	–

Error in H^1 norm blows up significantly as ϵ decreases.

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- SUPG provides a consistent way to add numerical diffusion.
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Future Work:

- Adaptive mesh refinement (Shishkin-type meshes).
- Comparison with Discontinuous Galerkin (DG) methods.

References

- H.-G. Roos, M. Stynes, and L. Tobiska, *Robust Numerical Methods for Singularly Perturbed Differential Equations: Convection-Diffusion -Reaction and Flow Problems*, 2nd ed., Springer, 2008.

- A. N. Brooks and T. J. R. Hughes, *Streamline Upwind/Petrov -Galerkin formulations for convection dominated flows with particular emphasis on the incompressible Navier-Stokes equations*, Comput. Methods Appl. Mech. Engrg., 32 (1982), pp. 199–259.

Thank you!